SMART INDIA HACKATHON 2026

JalRakshak

Your Water. Your Village. Your Voice.

- **Problem Statement ID –** SIH26255
- **Problem Statement Title –** AI/ML solution to monitor the functionality of FHTC in real time at Gram Panchayat and household levels, with water-quality alerts, citizen feedback and IMIS/Sujal Gaon ID sync
- **Theme –** Clean & Green Technology
- **PS Category –** Hardware
- **Team ID –** [enter from portal]
- **Team Name –** Team Vistaar



Team Vistaar

JalRakshak: Closed-Loop Water-Service Assurance

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98.1%
surveyed households in Har Ghar Jal villages had a tap

76.0%
assessed functionality status of household tap connections

A tap is not the same as a working water service.

JJM already has dashboards, WQMIS, Jal Seva Aankalan, Sujalam Bharat IDs and IoT pilots. We did not find a public system that links sensor warnings, citizen complaints and pipeline records into one prioritised repair that is verified after restoration.

Source: DDWS, Functionality Assessment of Household Tap Connection – National Report 2024 (weighted sample estimates) [1]

Proposed Solution

- **Sensor nodes** at pump, tank, pipeline head and tail – not at every house
- **Citizen app** (English / తెలుగు / हिंदी): report no, low or dirty water or leakage with a photo; track status
- **Pipeline records:** age, material, repairs, corrosion findings → explainable risk score
- **Gram Panchayat console:** evidence, maintenance tasks, restoration check

How it addresses the problem

- Detects supply failures continuously, between periodic surveys
- Separates a faulty sensor from a real supply fault
- Links each complaint to a pipe and zone, so the right asset is repaired first
- Closes an incident only when readings recover – measures service, not just assets

Innovation and uniqueness

- **Evidence fusion:** sensors + complaints + photos + pipe history + supply schedule → one explainable incident
- **Preventive maintenance:** inspect high-risk pipes before visible failure
- **Verified restoration,** not self-declared closure
- **Complements JJM systems;** Sujal Gaon / Sujalam Bharat-ready IDs (mock in prototype)

Predict → **Report** → **Review** → **Prioritise** → **Maintain** → **Verify** → **Inform**

Closed loop: early warning → citizen report → official review → prioritised task → repair → sensor-verified restoration → villager informed

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TECHNICAL APPROACH

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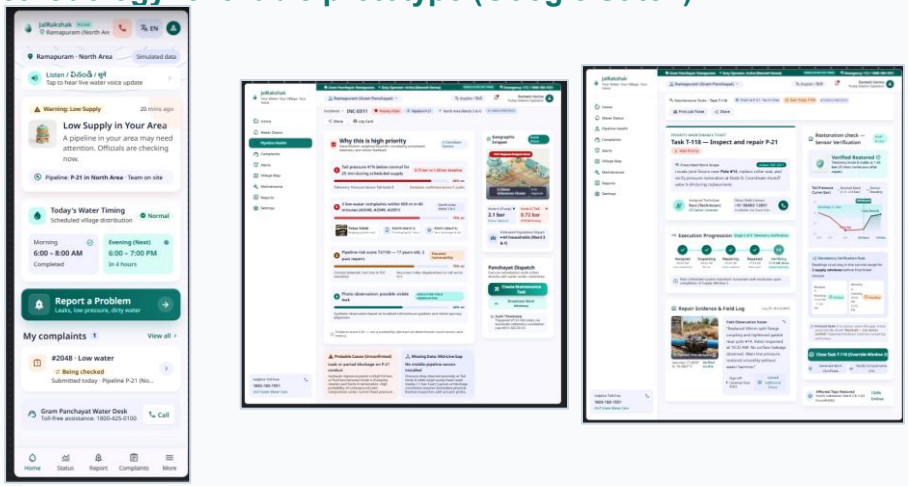
FIELD	Pump · current	Tank · level	Pipeline head · flow + pressure	Pipeline tail · flow + pressure	Citizens · photo, voice, location
EDGE	Node A: ESP32 + LoRa (pump/tank)	Node B: ESP32 + LoRa, solar (tail)	Raspberry Pi gateway · offline buffer	4G · MQTT	
CLOUD	FastAPI ingest + APIs	SQLite → PostgreSQL / PostGIS	Photo store · role-based access	Mock Sujal Gaon / Sujalam Bharat IDs	
AI / ML	Rules	Sensor-health gate	Anomaly: z-score → Isolation Forest	Pipeline risk score	Evidence fusion
ACTION	Incident + evidence	GP maintenance task	Repair	Verification window	Citizen notified

Build path: Google Stitch prototype → React PWA · sensor simulator → bench rig on the same ingest API · rules first, ML added as data grows

Technologies to be used

- **Hardware:** ESP32 + LoRa, hall-type flow ×2, pressure transducer ×2, ultrasonic level, CT pump-current sensor, Raspberry Pi gateway, solar + battery, IP65 enclosures
- **Software:** React + Vite + Tailwind (mobile-first PWA), Leaflet + OpenStreetMap, Python FastAPI, SQLite → PostgreSQL/PostGIS, MQTT (Mosquitto)
- **AI/ML:** Python, pandas, scikit-learn: rules, robust z-score, Isolation Forest trained on normal runs; explainable risk and fusion scores
- **Image triage:** simulated in prototype; a vision model only after labelled-data evaluation

Methodology: clickable prototype (Google Stitch)



Prototype screens use simulated data · link in reference [11]

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FEASIBILITY AND VIABILITY

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Worked example: Pipeline P-21 (illustrative, simulated data)

North Area · iron (record) · 17 years underground (record) · 3 past repairs · corrosion 67/100 (visual inspection of an exposed section)



Risk score 72 / 100 · HIGH · Inspect soon

Complaint #2048: "Low water" + photo

↓

Simulated image check: "possible visible leak – inspection recommended"

↓

Fused with tail pressure −41% for 25 min + 2 nearby complaints → HIGH incident

↓

GP assigns maintenance task → repair logged with photo

↓

Readings normal for 2 supply windows → Verified Restored → villager notified

The risk score ranks inspection priority; it is not a failure probability. A photo cannot show buried-pipe condition.

Analysis of the feasibility of the idea

- Commodity, student-buildable hardware; open-source software
- Works from day one with rules; ML added as real data accumulates
- A few nodes per scheme + citizens as observers: far cheaper than per-house meters
- JJM pilots already ran off-grid (solar + battery) sensor monitoring with RF + cellular links [7]

Potential challenges and risks → Strategies for overcoming these challenges

False alarms	Persistence windows, schedule-aware rules; HIGH needs ≥ 2 evidence types
Sensor fault or drift	Heartbeat, stuck/out-of-range checks; "node offline" ≠ "no water"
Low-cost sensor accuracy	Bench calibration; deployment-grade EM flow, 4–20 mA pressure
Water-quality limits	pH / TDS / turbidity are indicators only → FTK / lab confirmation
Rural connectivity	Gateway store-and-forward; replay when the network returns
Citizen privacy	Coarse location, role-based photo access, audit log
Government integration	Mock IDs + adapter; live IMIS / Sujalam Bharat sync needs authorisation

Prototype status (as of submission)

Clickable UI (Google Stitch)	IN PROGRESS
Sensor readings	SIMULATED · bench rig planned
Risk score, fusion, verification logic	DESIGNED · code after submission
Image analysis	SIMULATED · clearly labelled
IMIS / Sujal Gaon sync	NOT IMPLEMENTED · needs authorisation

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IMPACT AND BENEFITS

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Potential impact on the target audience

Villagers
Before: Complaint disappears; no update

After: Report in their own language with a photo; told when water is actually restored

Gram Panchayat / VWSC
Before: Learns of failures late; no priority order

After: Ranked inspection list with evidence; proof that a repair worked. Only 55.2% of villages had a VWSC and 58.1% skilled O&M manpower [1]

District / State officers
Before: Coverage numbers, not service

After: Repeat-failure hotspots, overdue repairs, verified resolution trends

Benefits of the solution (social, economic, environmental, etc.)

Social
Less time lost to outages; quality-indicator anomalies trigger earlier testing

Economic
Inspect the riskiest pipes first; fewer repeat repairs and wasted visits

Environmental
Earlier leak detection reduces water and pumping-energy loss

Governance
Verifiable service records aligned with JJM 2.0's service-delivery focus [4]

Scalability
- 1 village → cluster per gateway → block → district → state
- Keyed to service-area (Sujal Gaon) and scheme (Sujalam Bharat) identities [2]
- Stateless APIs; PostgreSQL/PostGIS; LoRaWAN for multi-gateway networks
- Rules transfer to new villages on day one; models retrain per region as data grows

Roadmap



Limitations: bench-scale validation only; no field data yet; risk weights are expert judgement until real failure records exist.

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RESEARCH AND REFERENCES

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Details / Links of the reference and research work

[1] DDWS – Functionality Assessment of Household Tap Connection, National Report 2024
jaljeevanmission.gov.in/sites/default/files/2025-12/FHTC_National%2…

[2] PIB, 13 Aug 2026 – 75% of rural habitations covered with Sujal Gaon IDs
pib.gov.in/PressReleasePage.aspx?PRID=2298959

[3] PIB, 30 Dec 2025 – Jal Seva Aankalan launched
pib.gov.in/PressReleasePage.aspx?PRID=2209798

[4] Cabinet, Mar 2026 – JJM extended to Dec 2028 (₹8.69 lakh crore, service-delivery focus)
pmindia.gov.in/en/news_updates/cabinet-approves-extension-of-jal-je…

[5] JJM Dashboard (coverage, GIS, IoT pilot view)
ejalshakti.gov.in/JJMReport/JJMIndia.aspx

[6] JJM-WQMIS (laboratory and FTK water-quality testing)
ejalshakti.gov.in/WQMIS/

[7] PIB, 31 Mar 2021 – JJM deploys sensor-based IoT devices (off-grid solar + battery pilot)
pib.gov.in/PressReleaseIframePage.aspx?PRID=1708701

[8] Jal Jeevan Samvad, Oct 2023 – IoT water-quality monitoring with complaint tracking
jaljeevanmission.gov.in/sites/default/files/2025-05/JalJeevanSamvad…

[9] Systematic review: ML for anomaly detection in smart water metering networks (2024)
sciencedirect.com/science/article/pii/S1474706524000160

[10] SIH 2026 problem statement SIH26255, Ministry of Jal Shakti
sih.gov.in/sih2026PS

[11] JalRakshak clickable prototype – Google Stitch (simulated data)
stitch.google.com/projects/5969310015179761668

How to read our claims

VERIFIED

Figures with a [n] citation, checked against official sources on 16 Oct 2026

PROPOSED

Architecture, risk scoring, evidence fusion and verification: our design

SIMULATED

Village, Pipeline P-21, Complaint #2048, sensor readings, image analysis and scores: demo data, labelled on screen

NOT YET IMPLEMENTED

Field deployment, trained vision model, IMIS / Sujalam Bharat / JJM API integration

No model accuracy, field results or supplier prices are claimed. pH, TDS and turbidity readings are indicators that trigger confirmatory testing, not a safety certification.